Chapter 8 Emerging Trend in Databases

8.1 NoSQL Databases

8.2 Object Oriented Database and ORM

8.3 Parallel and Distributed Databases

8.4 Data Warehouse and Data Mining

8.1 NoSQL Databases

- NoSQL is a type of database management system (DBMS) that is designed to handle and store large volumes of **unstructured and semi-structured data.**
- Unlike traditional relational databases that use tables with pre-defined schemas to store data,
- NoSQL databases use flexible data models that can adapt to changes in data structures and are capable of scaling horizontally to handle growing amounts of data.

8.1 NoSQL Databases

- The term NoSQL originally referred to “non-SQL” or “non-relational” databases, but the term has since evolved to mean “not only SQL,” as NoSQL databases have expanded to include a wide range of different database architectures and data models.
- NoSQL relies upon a softer model known as the **BASE model**. **BASE (Basically Available, Soft state, Eventual consistency).**
  - **Basically Available:** Guarantees the availability of the data . There will be a response to any request (can be failure too).
  - **Soft state:** The state of the system could change over time.
  - **Eventual consistency:** The system will eventually become consistent once it stops receiving input.

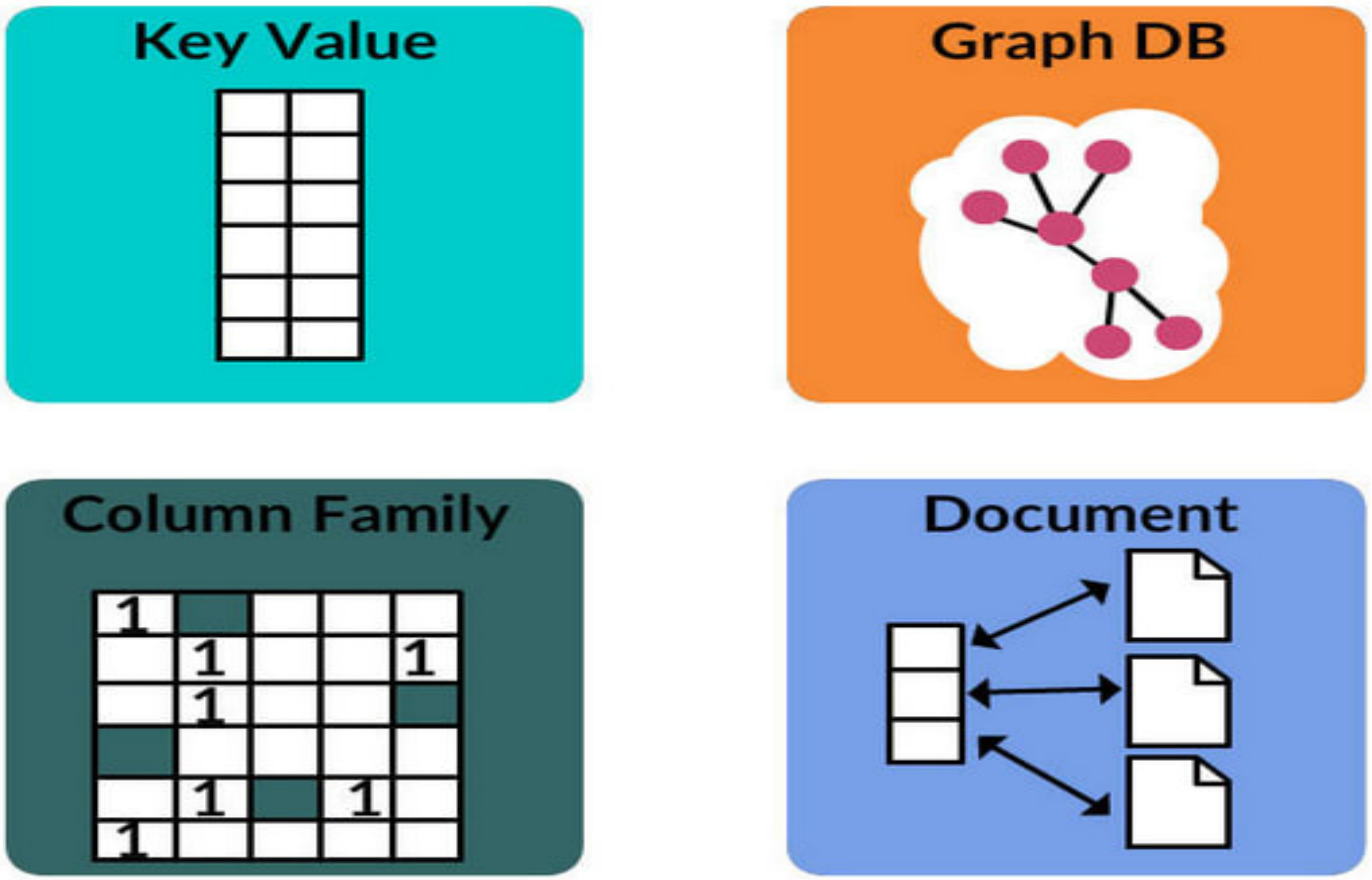
8.1 NoSQL Databases

Features/Characteristics

- ❖ Distributed computing system.
- ❖ Higher Scalability.
- ❖ Reduced Costs.
- ❖ Flexible schema design.
- ❖ Process unstructured and semi-structured data.
- ❖ No complex relationship.
- ❖ Open-sourced.

8.1 NoSQL Databases

Types(Categories) of NoSQL databases



8.1 NoSQL Databases

Types(Categories) of NoSQL databases

1. Key-Value Store Database:

- Termed to be the simplest form of NoSQL database of all other types, the key-value database is a database that stores data in a schema-less manner. This type of database stores data in the key-value format.
- Herein, a data point is categorized as a key to which a value (another data point) is allotted. For instance, a key data point can be termed as 'age' while the value data point can be termed as '45'.

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

1. Key-Value Store Database:

Key:1	ID:210		
Key:2	ID:411	Email: geeksforgeeks@gmail.com	
Key:3	UID:219	Name: Geek	Age:20

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

1. Key-Value Store Database:

Features

- **Scalability:** Large amounts of data and users.
- **Speed:** Large number of queries.
- **Data model:** Key-value pairs.
- **Consistency.**
- **Transactions.**
- **Querying ability.**

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

1. Key-Value Store Database:

Limitations

- No relationships among Multiple-Data.
- **Multi-operation Transactions:** If you are storing many keys and there is a failure to save one of the keys, you can't roll back the rest of the operations.
- **Query Data by 'value':** Searching the 'keys' based on some info found in the 'value' part of the key-value pairs.
- **Operation by groups:** As operations are confined to one key at a time, there exists no way to run several keys simultaneously.

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

1. Key-Value Store Database:

Real life examples

- ❖ Key-Value would be best-fit to store user profile:
  - ❖ userId, username.
- ❖ additional attributes/preferences:
  - ❖ Language
  - ❖ Country
  - ❖ Timezone
  - ❖ User favorites
  - ❖ and so on

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

1. Key-Value Store Database:

- DynamoDB
- Riak
- Berkeley DB
- Redis

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

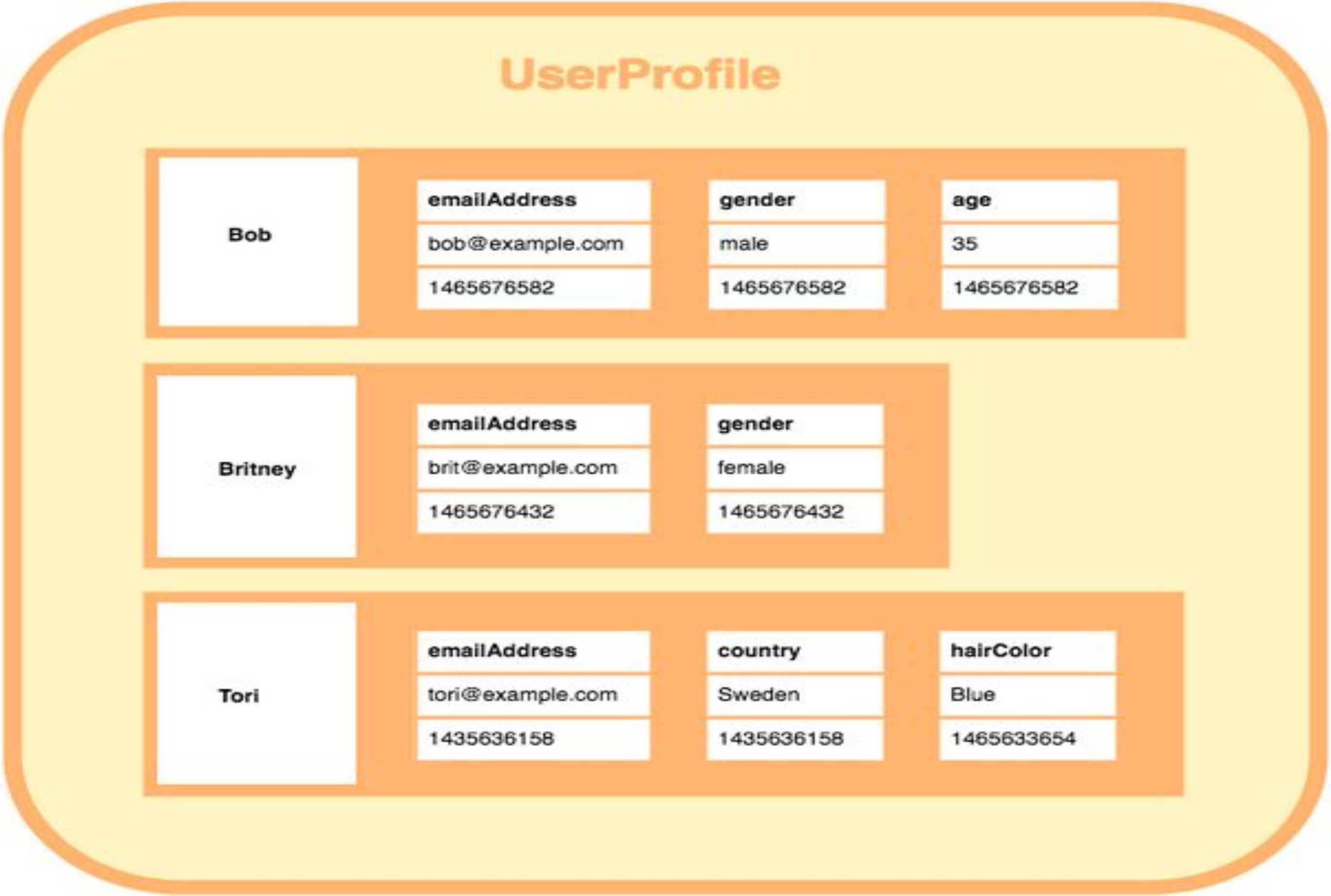
2. Column Store Database:

- ❖ Another type of NoSQL database is the column-oriented database. This type of database stores data in the form of columns that segregates information into homogenous categories.
- ❖ This allows the user to access only the desired data without having to retrieve unnecessary information.
- ❖ When it comes to data analytics in social media networking sites, the column-oriented database works very efficiently by showcasing data that is prevalent in the search results.

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

2. Column Store Database:



8.1 NoSQL Databases

Types(Categories) of NoSQL databases

2. Column Store Database:

Features

- **Compression:** Column stores are very efficient at data compression and/or partitioning.
- **Aggregation queries:** Due to their structure, columnar databases perform particularly well with aggregation queries (such as SUM, COUNT, AVG, etc).
- **Scalability:** Columnar databases are very scalable. They are well suited to massively parallel processing, which involves having data spread across a large cluster of machines – often thousands of machines.
- **Fast to load and query:** Columnar stores can be loaded extremely fast. A billion row table could be loaded within a few seconds.

Types(Categories) of NoSQL databases

2. Column Store Database:

Limitations

- ❖ **Incremental data loading:** It takes more time writing data than reading. Online Transaction Processing (OLTP) usage.
- ❖ **Queries against only a few rows:** Reading specific data takes more time than intended.

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

2. Column Store Database:

Real life examples

- ❖ A column family for vegetables of a supermarket.
- ❖ A column family for clients.
- ❖ A column family for users.

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

2. Column Store Database:

- Cassandra
- HBase
- Kudu

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

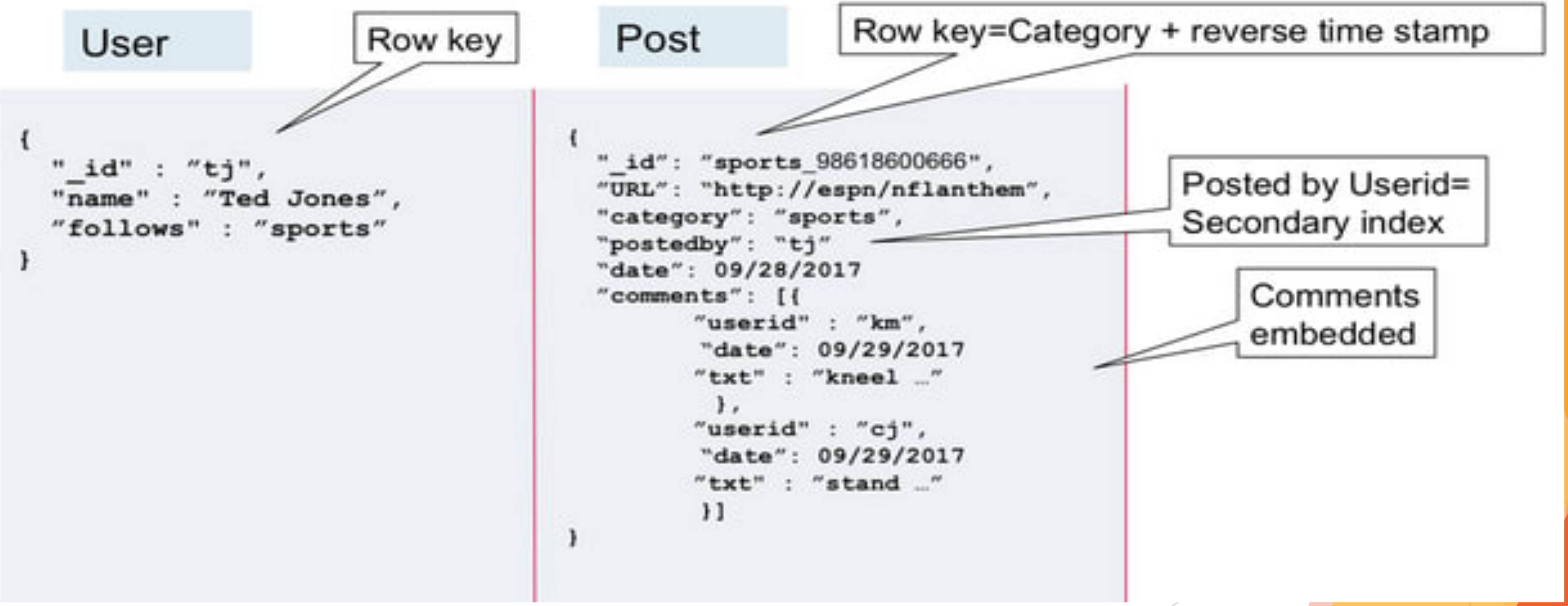
3. Document Database:

- The document database fetches and accumulates data in form of key-value pairs but here, the values are called as Documents.
- Document can be stated as a complex data structure. Document here can be a form of text, arrays, strings, JSON, XML or any such format.
- The use of nested documents is also very common. It is very effective as most of the data created is usually in form of JSONs and is unstructured.

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

3. Document Database:

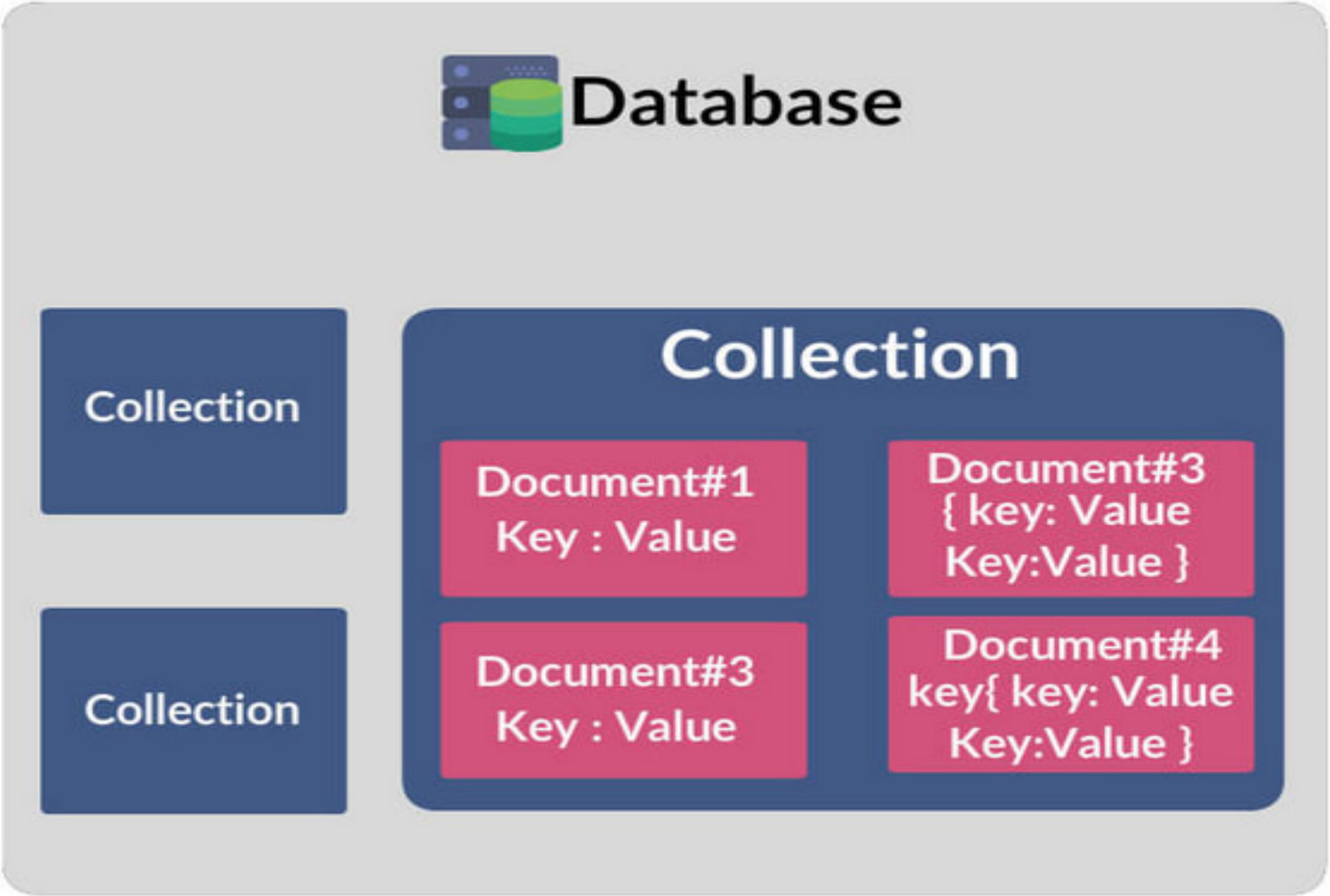


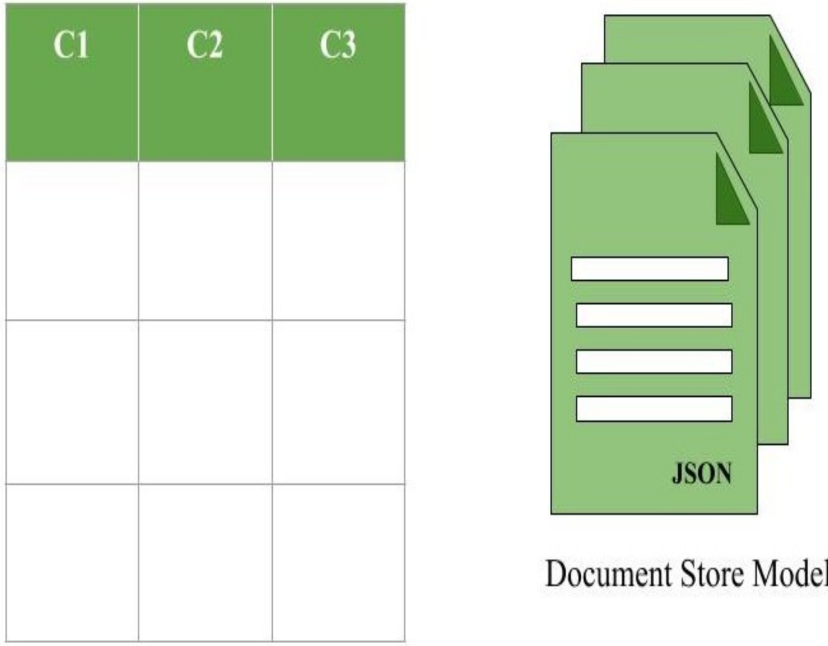
JSON Databases

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

3. Document Database:





Relational Data Model

Document Store Model

Data modelling

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

3. Document Database:

Features

- **Scalability:** For more complex objects.
- **Data model:** Collection of documents.
  - Similar to JSON and XML.
- Implements ACID transactions and adapt RDBMS characteristics.
  - Allows indexing of documents based on its primary identifier and properties.
  - Supports Query transactions (to an extent).
- Design pattern allows retrieving info in a single operation.
- Avoids performing joins within the application.

Types(Categories) of NoSQL databases

3. Document Database:

Limitations

- Base info duplication across multiple documents
- Complicates design resulting in inconsistency.
  - **Solution:** Link multiple documents using document identifiers (resembles foreign key in RDBMS)

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

3. Document Database:

Real life examples

- ❖ Shutterfly is a popular internet-based photo sharing and personal publishing company that manages a store of more than 6 billion images with a transaction rate of up to 8,000 operations per second.
- ❖ eBay has a number of projects running on MongoDB for search suggestions, metadata storage, cloud management and merchandizing categorization.

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

3. Document Database:

- MongoDB
- CouchDB

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

4. Graph Database:

- Graph Based Data Model in NoSQL is a type of Data Model which tries to focus on building the relationship between data elements.
- Graphs are basically structures that depict connections between two or more objects in some data.
- This pattern is very commonly used in social networks where there are a large number of entities and each entity has one or many characteristics which are connected by edges.

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

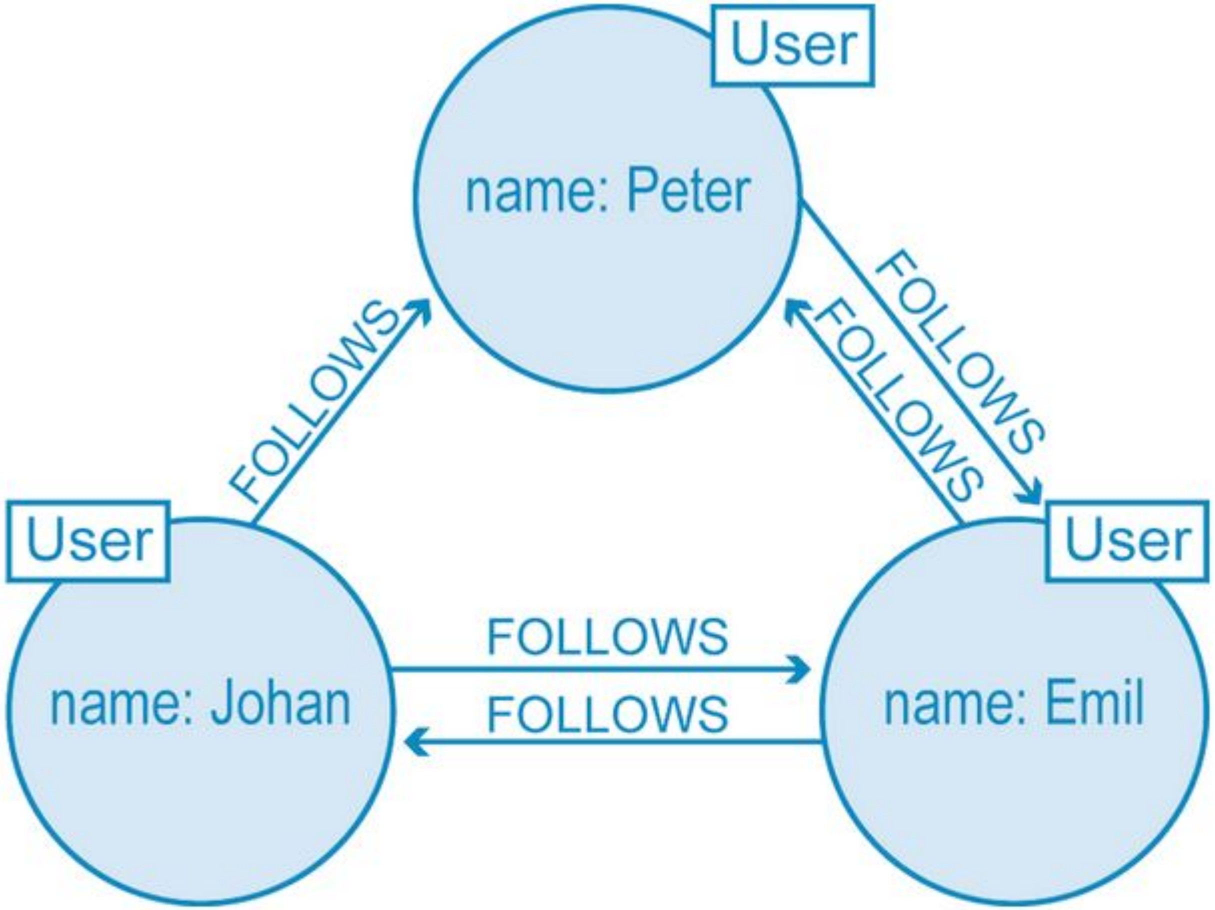
4. Graph Database:

Parts of a graph

- **Nodes -** representation of entities.
- **Properties -** Information about nodes.
  - Can be indexed.
- **Edges -** Relationships between nodes.
  - Can be indexed.
  - Unidirectional or bidirectional, no limit of edges.

Types(Categories) of NoSQL databases

4. Graph Database:



Types(Categories) of NoSQL databases

4. Graph Database:

Features

- ❖ Scales to the complexity of data.
- ❖ Focus on interconnectivity.
- ❖ Many query languages.

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

4. Graph Database:

Limitations

- **Lack of high performance concurrency:** In many cases, Graph Databases provide multiple reader and single writer type of transactions, which hinders their concurrency and performance as a consequence, somewhat limiting the threaded parallelism.
- **Lack of standard languages:** The lack of a well established and standard declarative language is being a problem nowadays. Neo4j is proposing Cypher and Oracle is working on a language.
- **Lack of parallelism:** One important issue is the fact that partitioning a graph is a problem. Thus, most do not provide shared nothing parallel queries on very large graphs. Thus, allowing for parallelism is intrinsically a problem.

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

4. Graph Database:

Real life examples

Graph Store is used to model all kind of different scenarios such as:

- Construction of a space rocket.
- Transportation system (roads and trains).
- Supply-chain and Logistics.
- Medical history.
- Fraud Detection.
- Network and IT Operations.

8.1 NoSQL Databases

Types(Categories) of NoSQL databases

4. Graph Database:

- ❖ Neo4J
- ❖ Giraph.

8.1 NoSQL Databases

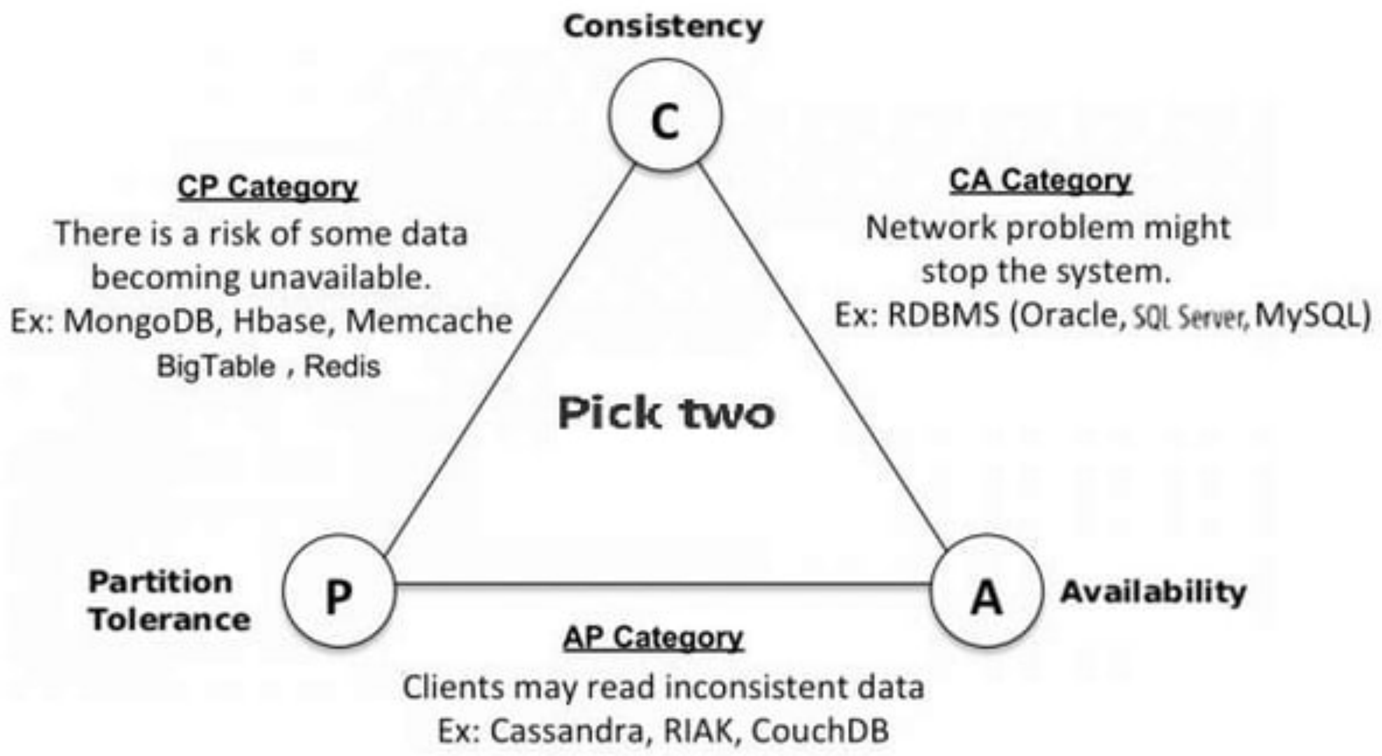
The CAP Theorem

- It is very important to understand the limitations of NoSQL database. NoSQL can not provide consistency and high availability together. This was first expressed by Eric Brewer in CAP Theorem.
- **CAP theorem or Eric Brewers theorem** states that we can only achieve at most two out of three guarantees for a database: Consistency, Availability and Partition Tolerance.

The CAP Theorem

- **Consistency:** Every read receives the most recent write or an error.
- **Availability:** Every request receives a (non-error) response – without the guarantee that it contains the most recent write.
- **Partition tolerance:** Even if there is a network outage in the data center and some of the computers are unreachable, still the system continues to perform.

The CAP Theorem



No system can provide more than 2 guarantees. In the case of a distributed systems, the partitioning of the network is a must, so the trade-off is always between consistency and availability.

Assignment

- Advantages and Disadvantages of NoSQL
- NoSQL vs SQL

8.2 Object Oriented Database and ORM

- **ODBMS** stands for Object-Oriented Database Management System, which is a type of database management system that is designed to store and manage object-oriented data.
- Object-oriented data is data that is represented using objects, which encapsulate data and behavior into a single entity.
- Object oriented data model is based upon real world situations.
- All these object have multiple relationships between them.

8.2 Object Oriented Database and ORM

- ❖ It maintains correspondence between real world and database objects without loosing integrity and identity.
- ❖ Object has state and behavior
- ❖ Entity is represented as a class.
- ❖ Instances of the class are objects.
- ❖ Within an object , the class attributes takes specific values.

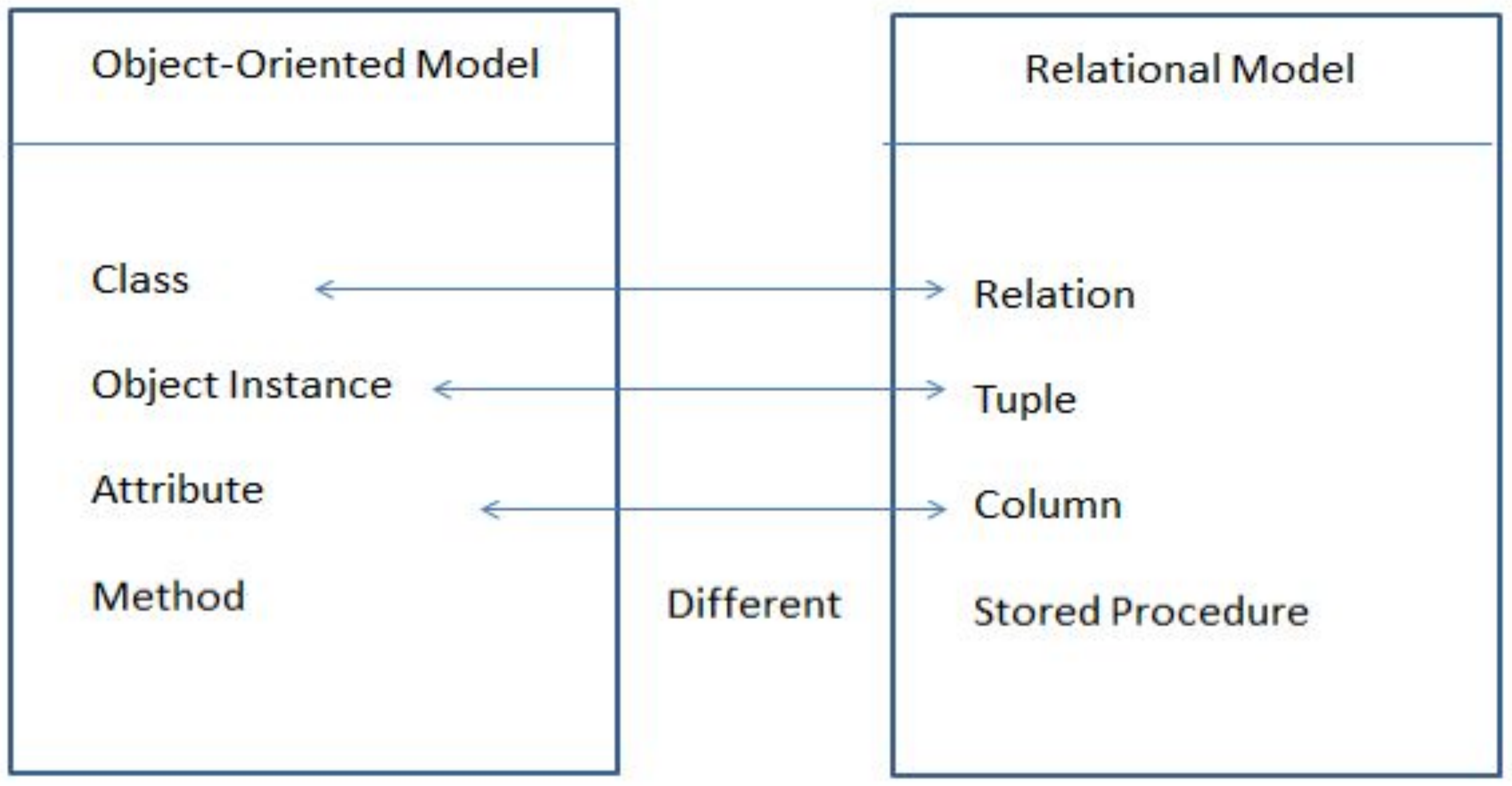
8.2 Object Oriented Database and ORM

Elements of object oriented model

- ❖ Object
- ❖ Attributes
- ❖ Methods
- ❖ Class

8.2 Object Oriented Database and ORM

Object v/s Relational Model



8.2 Object Oriented Database and ORM

Feature of object oriented Database

- **Object-Oriented Data Model:** OODBs use an object-oriented data model to represent data as objects, which consist of attributes and methods.
- **Encapsulation:** OODBs support encapsulation, which means that an object's data and methods are bundled together, providing data security and integrity.
- **Inheritance:** OODBs allow objects to inherit attributes and methods from other objects, enabling the creation of hierarchical relationships and code reuse.

8.2 Object Oriented Database and ORM

Feature of object oriented Database

- **Polymorphism:** OODBs support polymorphism, allowing objects to be treated as instances of their parent class, enabling flexibility and extensibility.
- **Complex Data Types:** OODBs support complex data types, such as arrays, lists, and sets, providing flexibility in data representation.
- **Query Language:** OODBs provide a query language that allows users to retrieve and manipulate objects using a language tailored for object-oriented data.

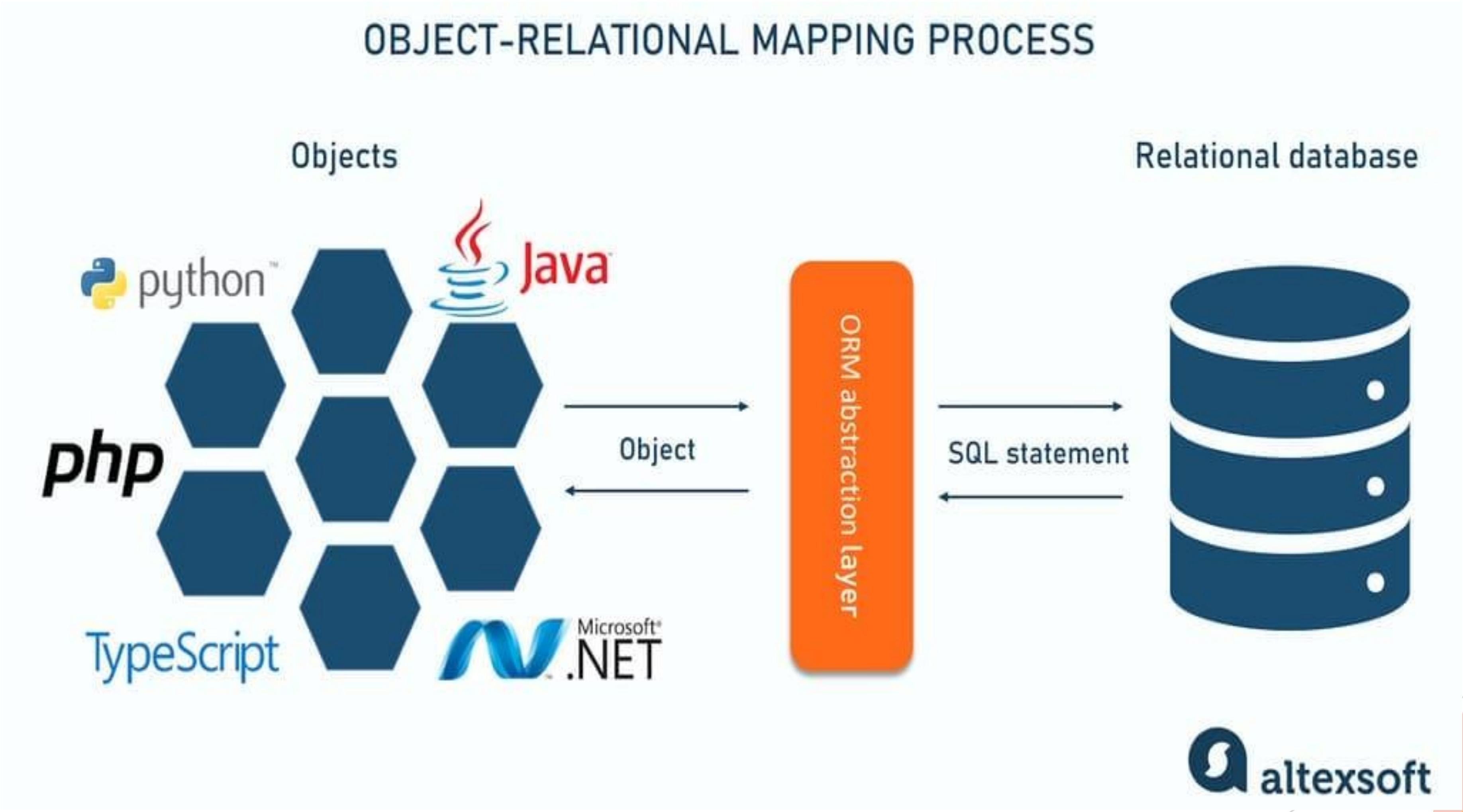
8.2 Object Oriented Database and ORM

- **Object Relational Mapping (ORM)** is a technique used in creating a "bridge" between object-oriented programs and, in most cases, relational databases.
- The idea of ORM is **based on abstraction**. The ORM mechanism makes it possible to address, access and manipulate objects without having to consider how those objects relate to their data sources.

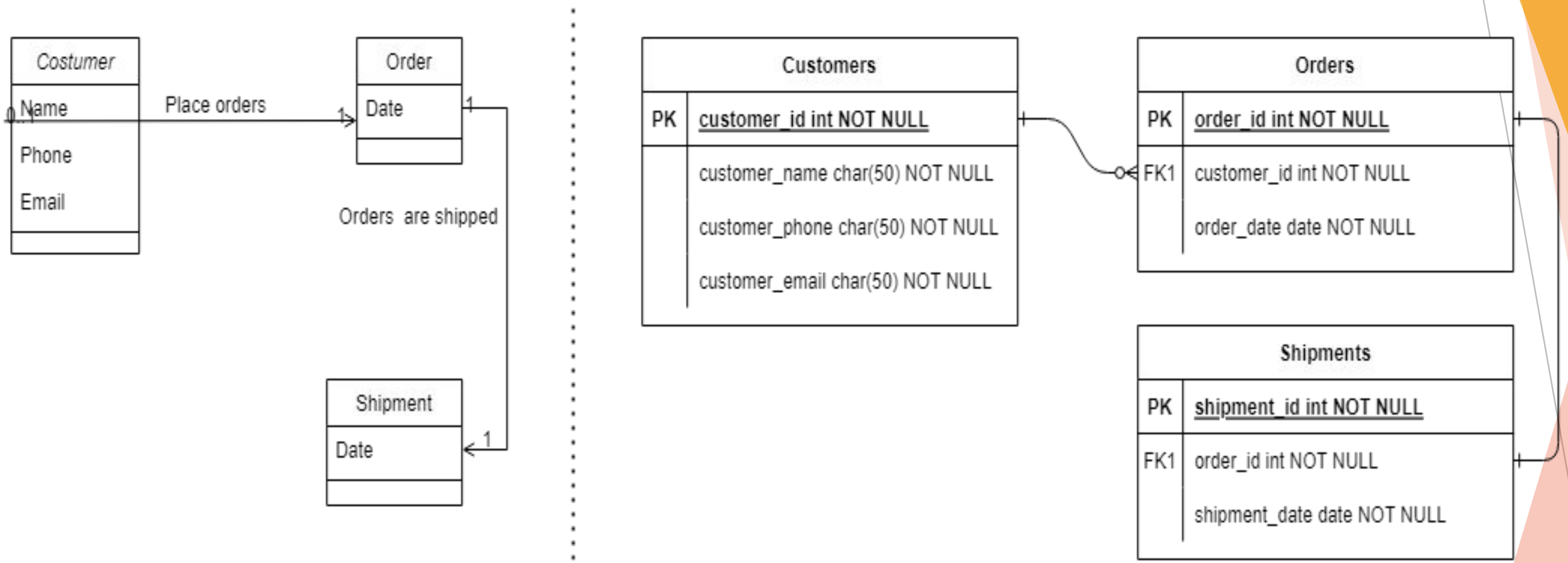
8.2 Object Oriented Database and ORM

- **ORM** is a programming technique for converting data between incompatible type systems into object oriented programming language.
- This creates a virtual object database , which can be used from within the programming language.
- It is necessary because objects are almost non-scales values while the DBMS can only store and manipulate scalar values organized within tables.
- ORM convert object values into group of singular values for storages in the database.

8.2 Object Oriented Database and ORM



8.2 Object Oriented Database and ORM



Model Classes **ORM** **Model's mapped tables**

8.2 Object Oriented Database and ORM

Here's an example of SQL code that retrieves information about a particular user from a database:

SELECT id, name, email, country, phone_number FROM users WHERE id = 20;

On the other hand, an ORM tool can do the same query as above with simpler methods. That is:

users.GetById(20);

So the code above does the same as the SQL query. Note that every ORM tool is built differently so the methods are never the same, but the general purpose is similar.

8.2 Object Oriented Database and ORM

Popular ORM Tools for Java

- Hibernate, Apache OpenJPA, EclipseLink

Popular ORM Tools for Python

- Django, Web2py, SQLObject

Popular ORM Tools for PHP

- Laravel, CakePHP, Doctrine

Popular ORM Tools for .NET

- Entity Framework, Nhibernate, Dapper

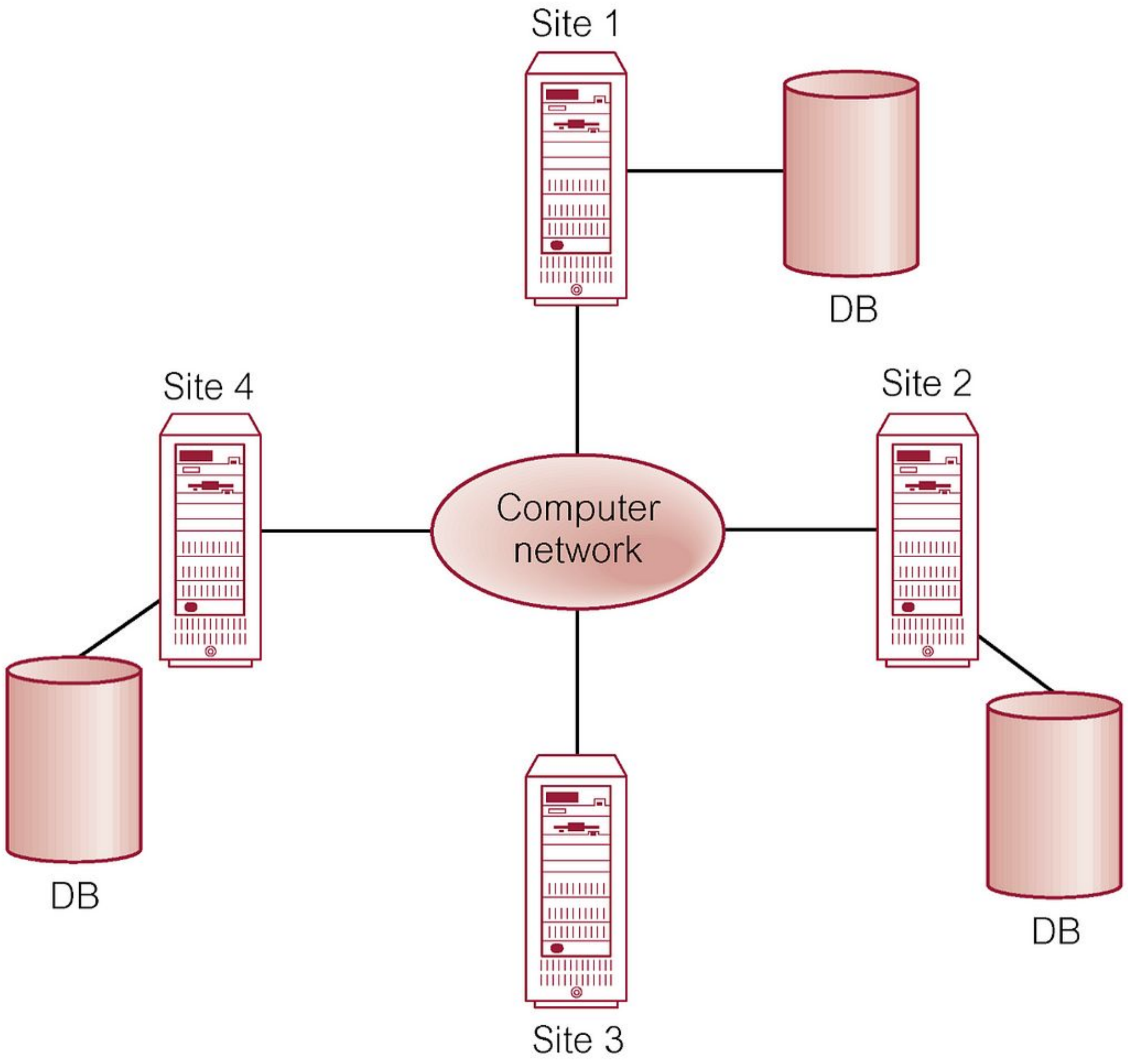
Assignment

- Advantages and Disadvantages of ODBMS
- Advantages and Disadvantages of ORM
- ODBMS vs RDBMS

8.3 Distributed Database

- The database is stored on several computers known as sites.
- Each site is largely coupled.
- There is no any stored physical component.
- Communication between sites is done with network.
- Local transaction access data only from site where it was initiated.
- Global transaction access data from different sites.

8.3 Distributed Database



8.3.1 Heterogeneous Database

- Each site uses different schema.
- Difficulty in query and transaction processing.
- Provide limited facility for cooperation
- Also, a particular site might be completely unaware of the other sites.
- Different computers may use a different operating system, different database application.

8.3.2 Homogeneous Database

- All sites have identical dbms software
- The operating system, database management system and the data structures used – all are same at all sites.
- Agree to cooperate in processing user requests.
- Each site surrender part of anatomy as a right to change schema
- It appears as a single system for users.

8.3.3 Data replication

- Approach to store distributed data in which system contains multiple copies of data in different sites.
- Full replication indicates storing at all sites,
- Fully redundant indicates each site having entire copy of database.
- It ensures availability ,parallelism and reduced data transfer
- Update cost is high
- Difficult for concurrency control

8.3.4 Data fragmentation

- ❖ Approach to store distributed in which relation r is divided.
- ❖ Fragments with sufficient information to reconstruct r
- ❖ Each schema should have common candidate key to ensure lossless join.
- ❖ In horizontal each tuple of r is assigned to one or more fragments.
- ❖ It allows parallel processing and easy access.
- ❖ In vertical schema is split into several smaller schemes.
- ❖ It allows parallel processing on relation.

8.4 Parallel Database

A Parallel Database is a database that is built up of multiple processors and storage units, located in one physical environment or they belong to a closely coupled system and are involved in performing the database tasks in parallel.

Single processor systems can no longer meet-requirements of scalability and performance and also the same throughput can be achieved as single, larger machine by replacing with multiple smaller machines.

8.4 Parallel Database

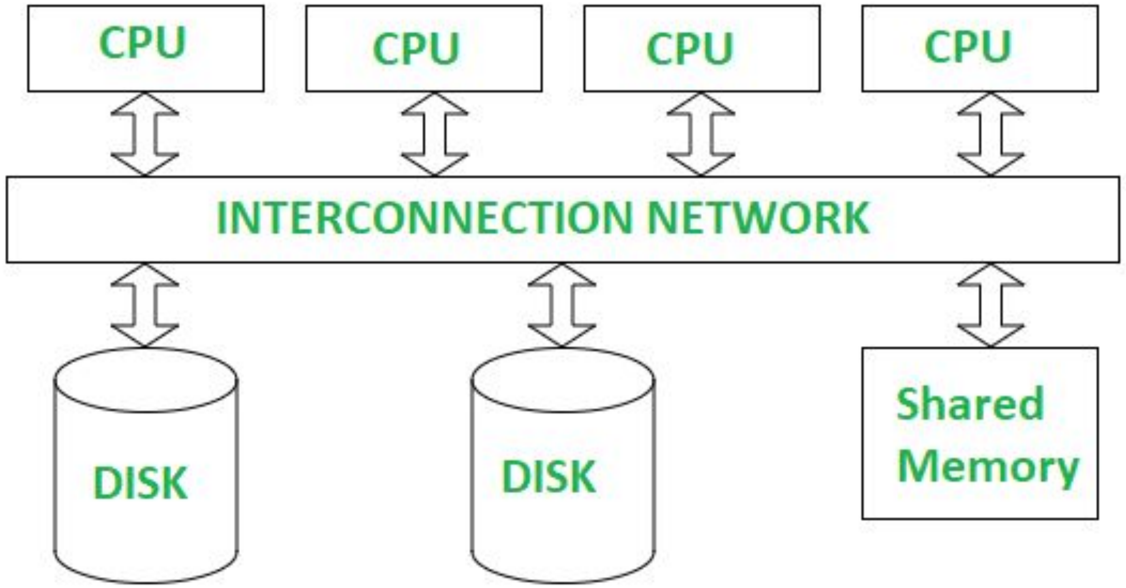
Parallel Databases can be broken down into 3 categories

- Shared Memory
- Shared Disk
- Shared Nothing

8.4 Parallel Database

1. Shared Memory

Multiple processors have direct access to single, shared memory space, this means all processors work with the same memory.

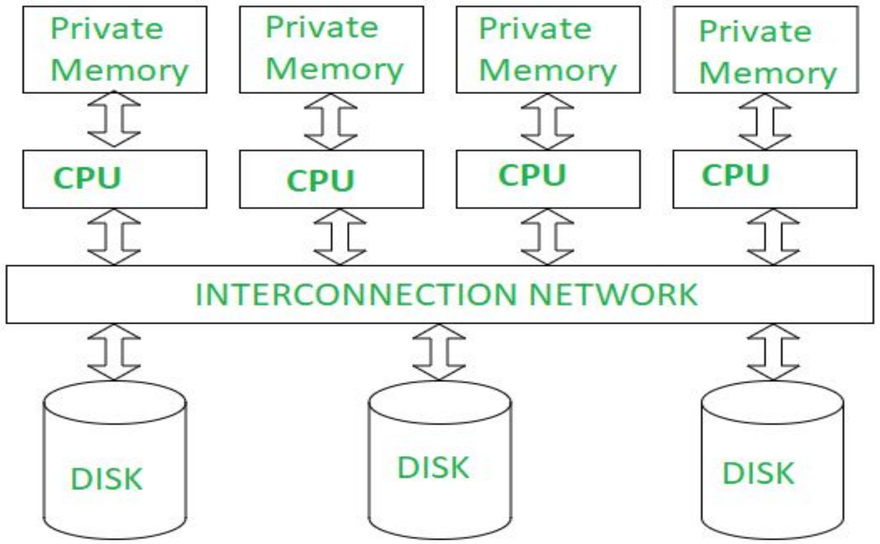


How It Works*: Imagine each person has their own notebook and their own filing cabinet, and they work independently. If they need to share information, they communicate over a network.*

8.4 Parallel Database

2. Shared Disk

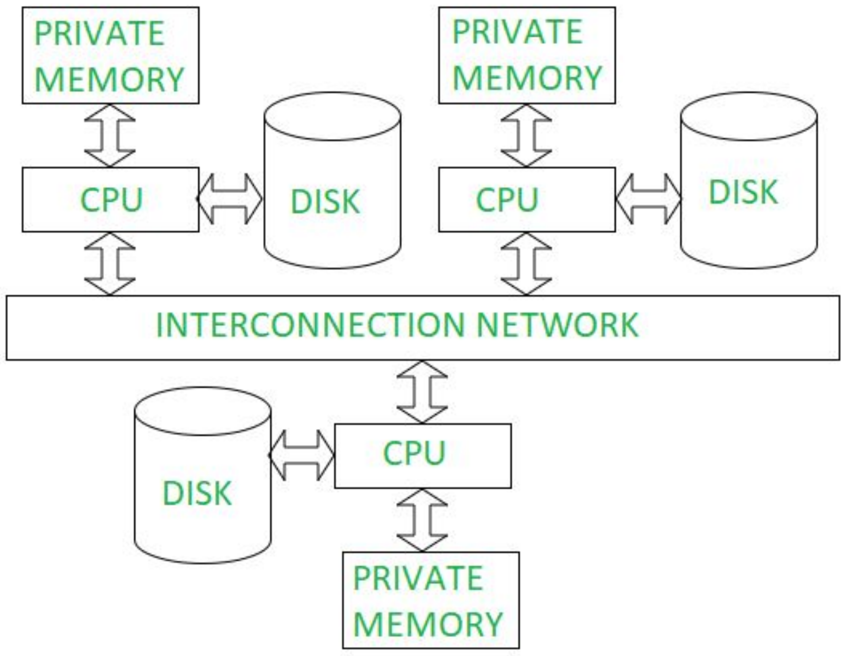
In a Shared Disk architecture, each processor has its own memory, but all processors share the same disk (or storage). Processors can read/write to the same disk, but have separate memories. Remember that each node has it's own Operation System.



How It Works: *Imagine several people with their own notebooks (memory) working together, but they all write to and read from a shared filing cabinet (disk).*

3. Shared Nothing

In a Shared Nothing architecture, each processor has its own memory and its own disk. The processors work independently, and they communicate only when needed. There's no shared memory/disk, so each processor manages its own data.

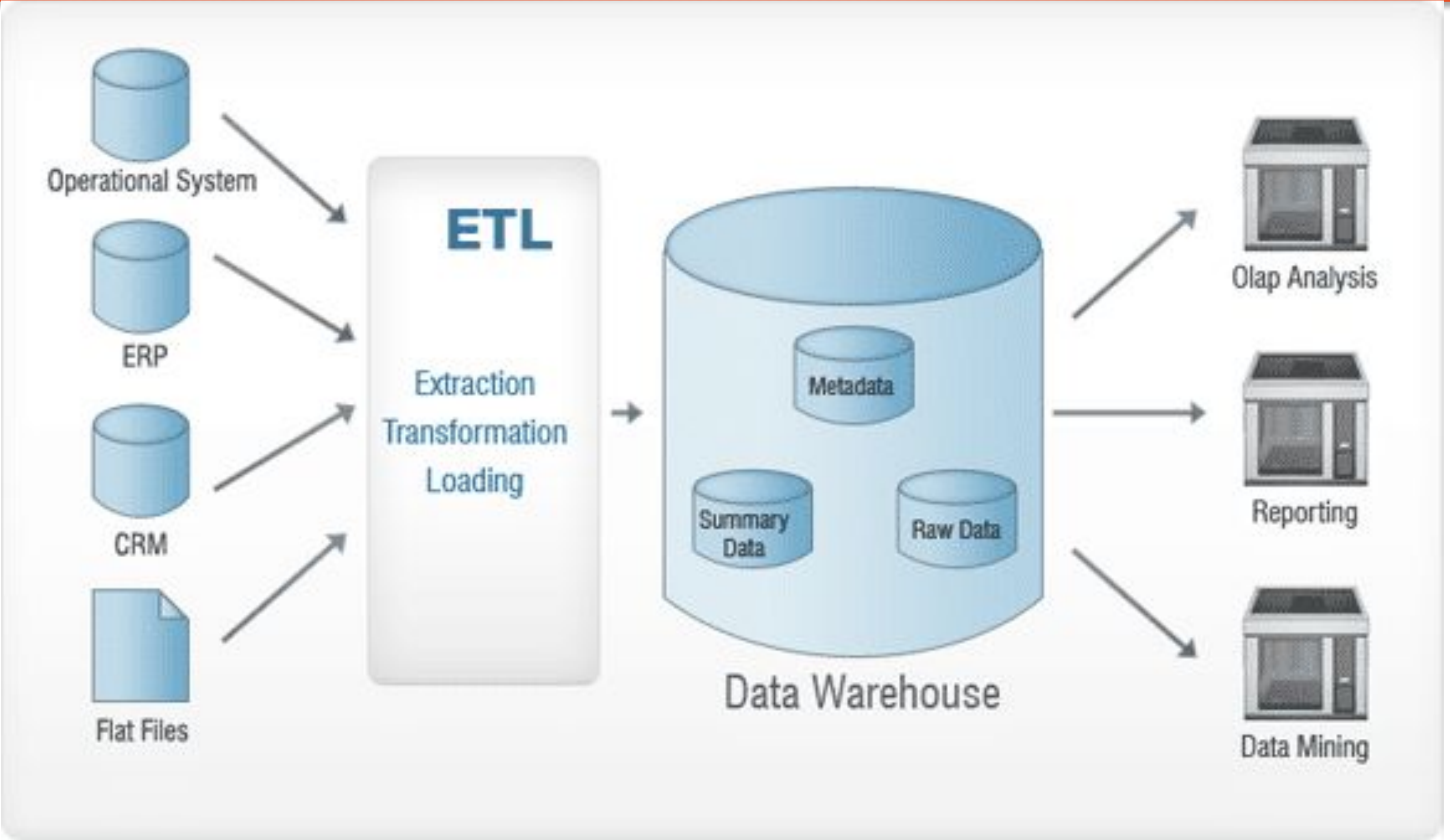


How It Works*: Imagine several people with their own notebooks (memory) working together, but they all write to and read from a shared filing cabinet (disk).*

8.4 Data Warehouse

- Data warehouse is a single ,complete and consistent data store obtained from a variety of sources made available to end users in the understandable form and used as a business context.
- It is subject oriented, integrated , time-variant and non-volatile collection of data to support decision making process.

8.4 Data Warehouse



- **Extraction**

Extraction is essentially done to collect all the required data from the source systems using as few resources as possible.

8.4 Data Warehouse

- **Transformation**

The main aim is to insert the extracted data into the database in a general format. This is because different sources will have different formats of storing the data – for example, one data source might have data in “dd/mm/yyyy” format, and the other might have it in “dd-mm-yy” format.

- **Loading**

In this step, the transformed data is loaded into the target database.

Online Analytical Processing (OLAP)

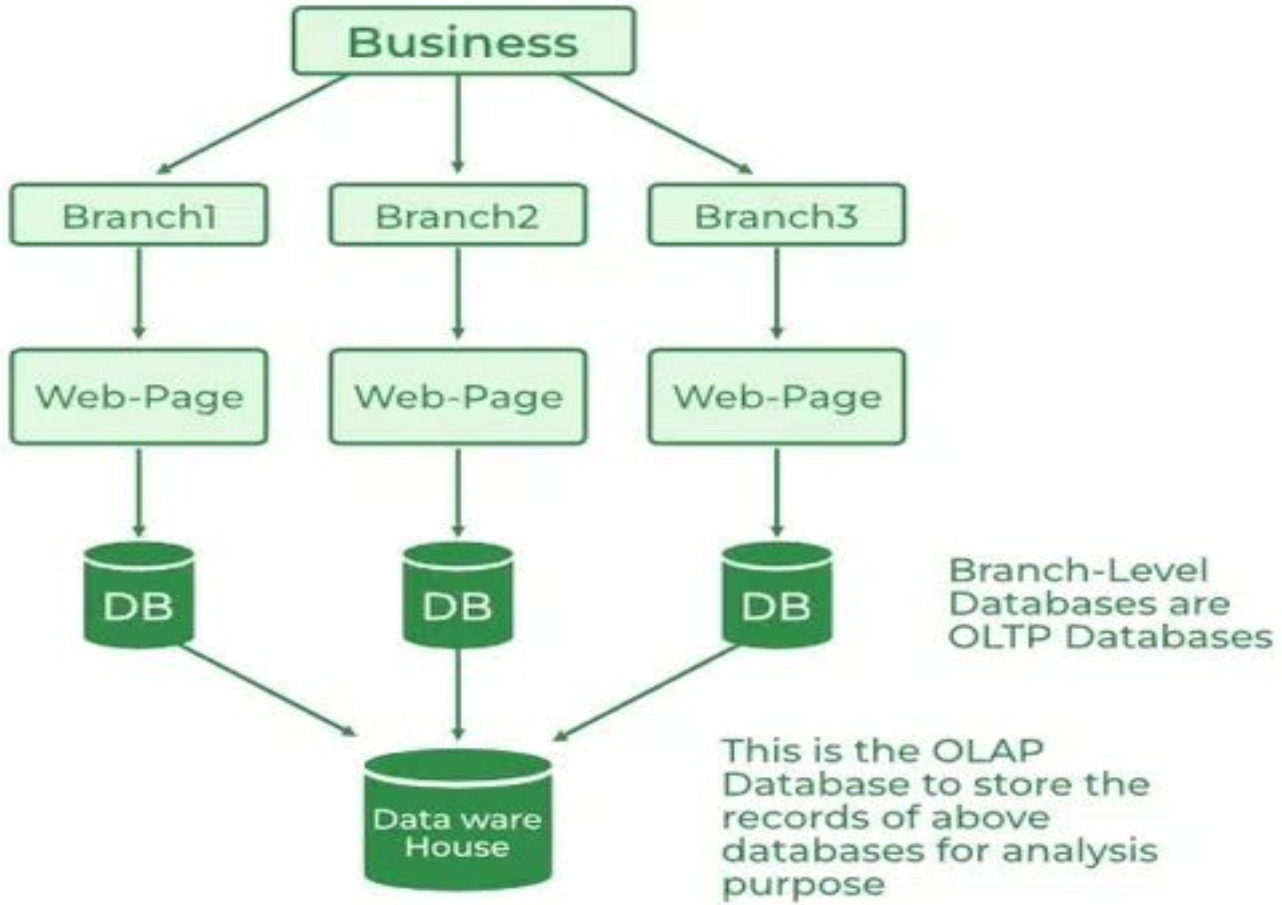
- OLAP systems have the capability to analyze database information of multiple systems at the current time.
- The primary goal of OLAP Service is data analysis and not data processing.
- Online Analytical Processing (OLAP) consists of a type of software tool that is used for data analysis for business decisions.
- OLAP provides an environment to get insights from the database retrieved from multiple database systems at one time.

Online Analytical Processing (OLAP)

OLAP Examples

- ❖ Any type of Data Warehouse System is an OLAP system.
- ❖ The uses of the OLAP System are described below.
  - ❖ Spotify analyzed songs by users to come up with a personalized homepage of their songs and playlist.
  - ❖ Netflix movie recommendation system.

Online Analytical Processing (OLAP)



Online Transaction Processing (OLTP)

- ❖ OLTP has the work to administer day-to-day transactions in any organization.
- ❖ The main goal of OLTP is data processing not data analysis.
- ❖ Online transaction processing provides transaction-oriented applications in a 3-tier architecture.
- ❖ OLTP administers the day-to-day transactions of an organization.

Online Transaction Processing (OLTP)

OLTP Examples

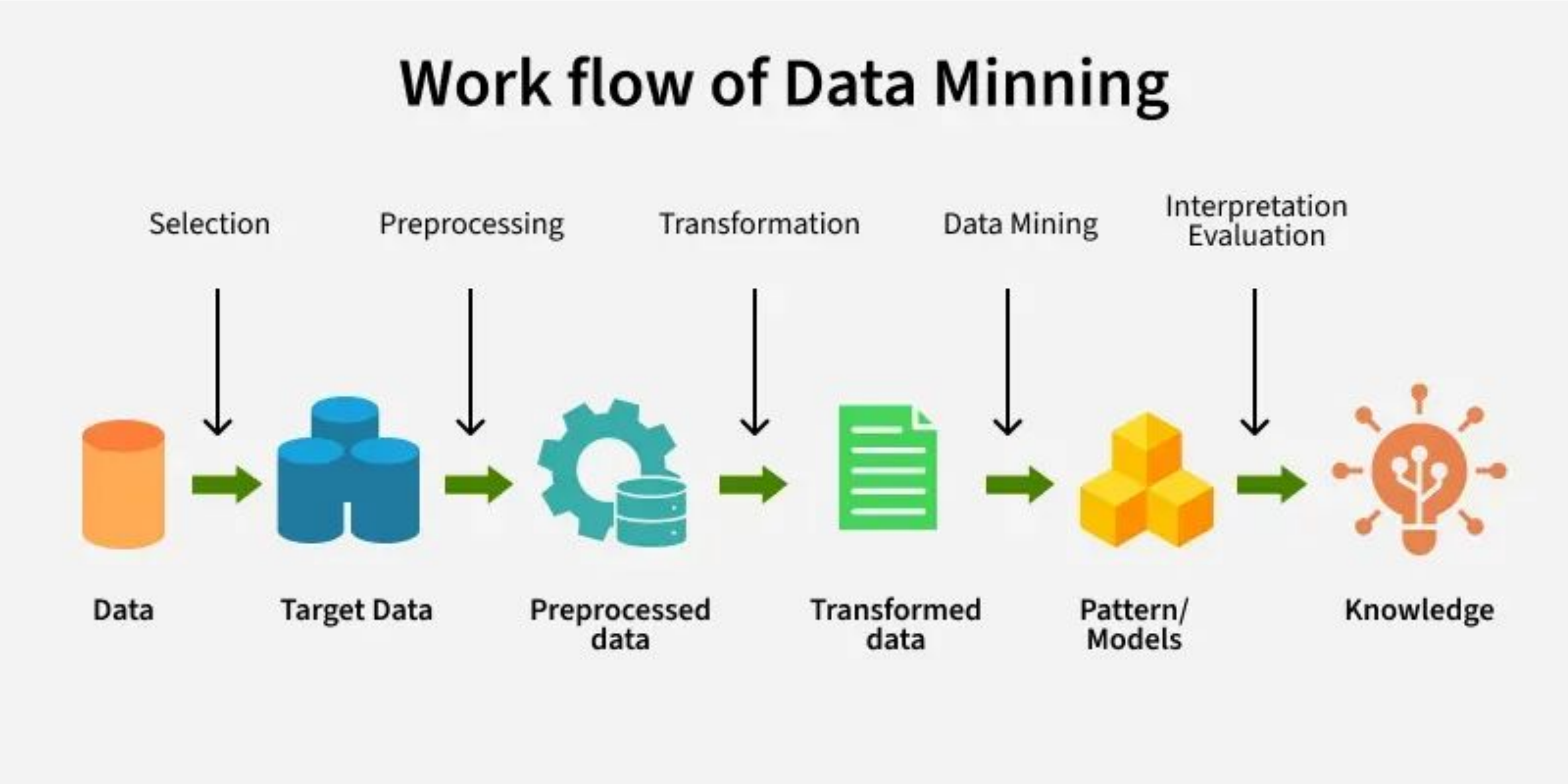
- ❖ An example considered for OLTP System is ATM Center a person who authenticates first will receive the amount first and the condition is that the amount to be withdrawn must be present in the ATM.
- ❖ The uses of the OLTP System are described below.
  - ❖ ATM center is an OLTP application.
  - ❖ OLTP handles the ACID properties during data transactions via the application.
  - ❖ It’s also used for Online banking, Online airline ticket booking, sending a text message, add a book to the shopping cart.

8.4 Data Mining

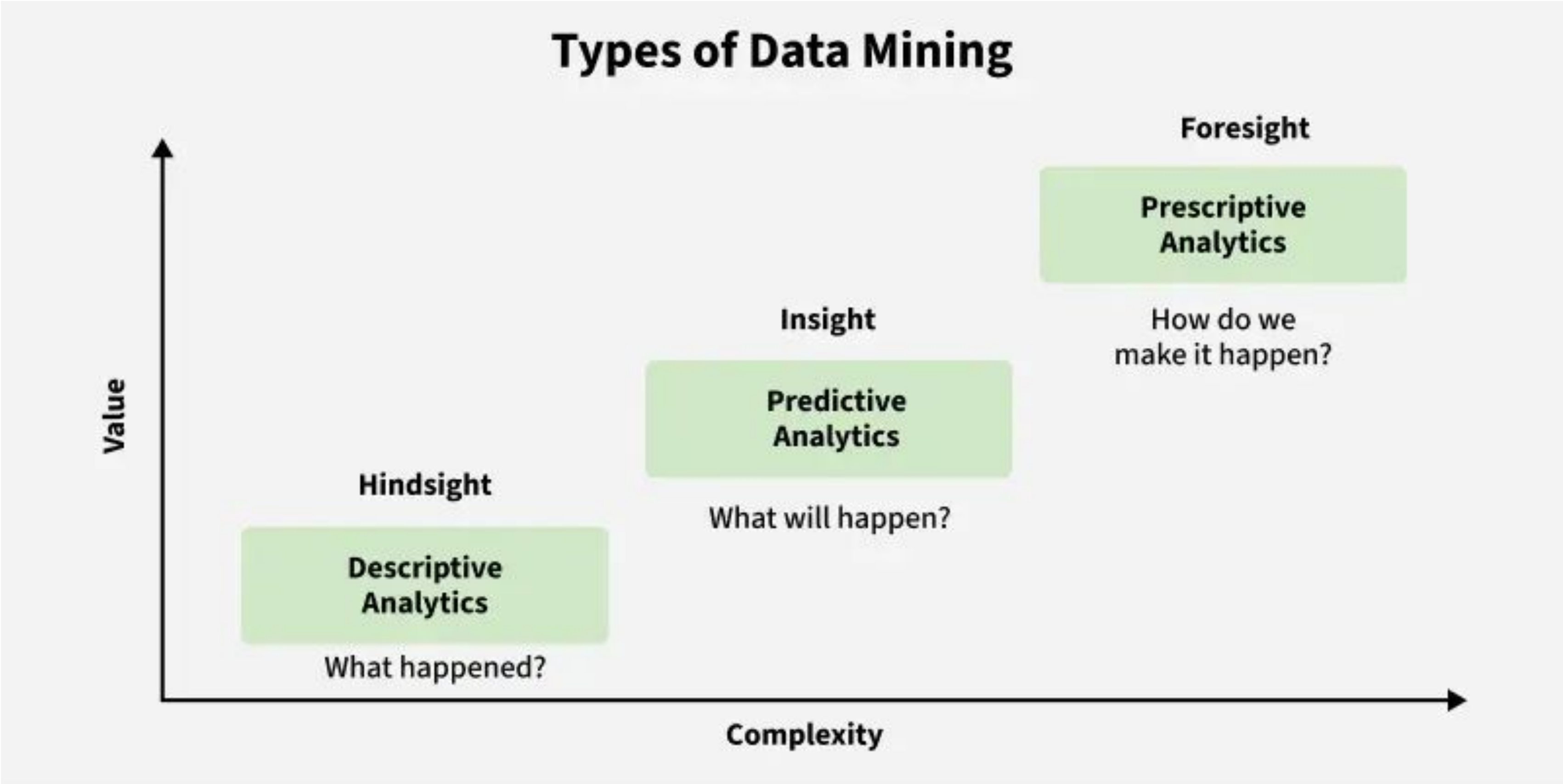
- In the process of data mining, data is carefully extracted and analysed to fetch nothing other than useful information. Here, all the hidden patterns are researched from the dataset, thereby predicting future behaviour. Besides, it can seamlessly indicate and discover unique relationships through the data.
- On the other hand, data mining makes the best use of artificial intelligence, statistics, machine learning systems, databases, etc. It is used for figuring out the hidden patterns within the data. Additionally, it also supports all business-related queries that take loads of time to resolve.

8.4 Data Mining

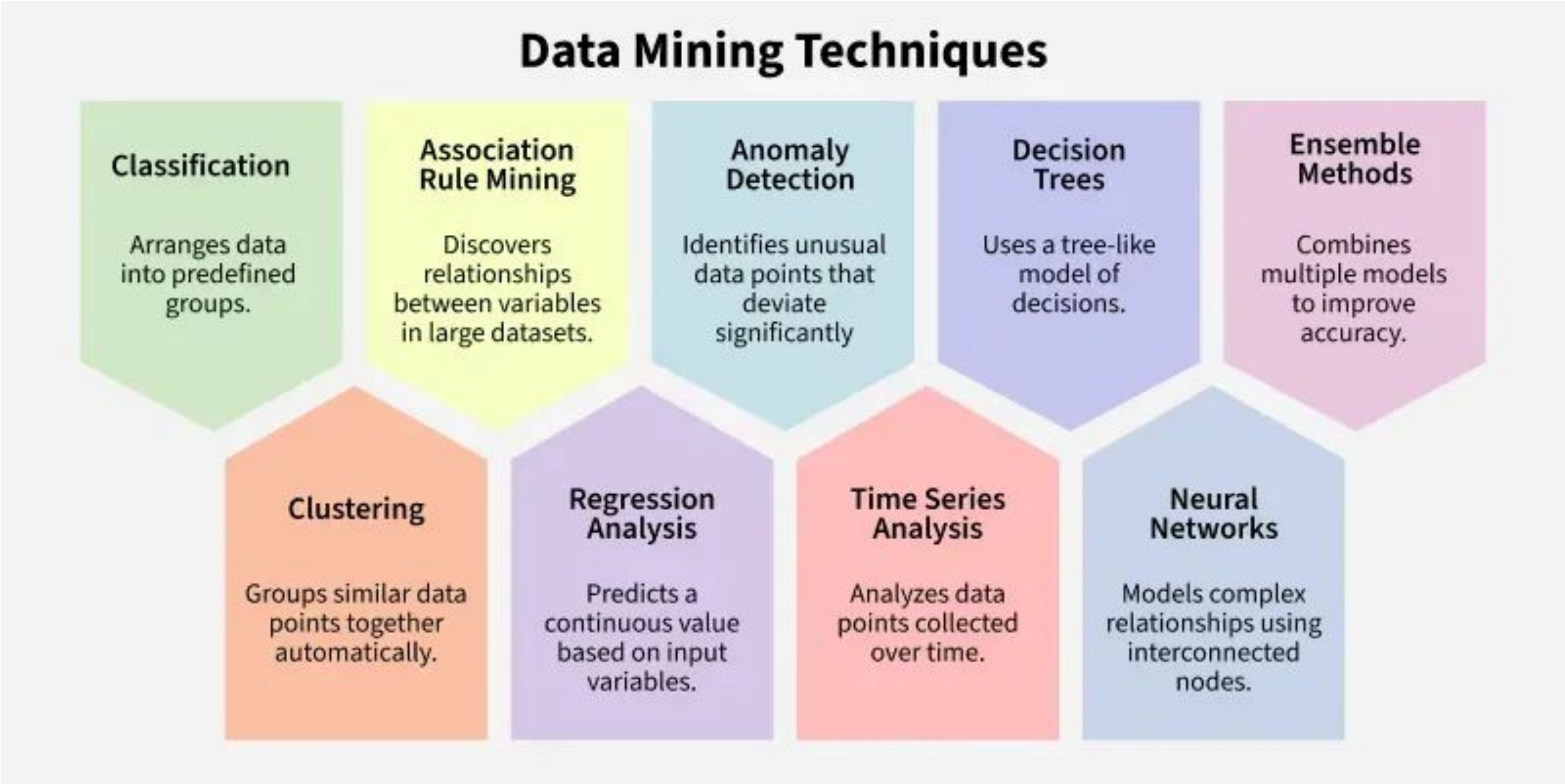
8.4 Data Mining



8.4 Data Mining



8.4 Data Mining



Assignment

- ❖ Advantages and Disadvantages of Parallel and Distributed Database
- ❖ Distributed v/s Parallel Processing
- ❖ Advantages and disadvantages of data warehouse
- ❖ OLAP v/s OLTP

This is the end of the lecture!
I hope you enjoyed it.
Thank You